Negative results and claim revision

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Preserve negative results without turning a tie into equivalence or selected demonstrations into confirmation.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

Robustness across conditions

My remembered idea, example and correction

Running the primary experiment

My remembered idea, example and correction

Pilot experiments

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Negative results state that evidence did not meet a declared expectation, with the original method and limitations retained. Main prediction expects B support strictly above A; both pass 8/10 = 80%, so observed strict improvement is absent. This does not prove equal population performance or universal uselessness. Separate picture predicts higher B correctness but both score 6/8 = 75%.

  Zero observed count difference is not an equivalence test, and individual correct/error identities are unknown. A new task chosen because A fails, on which B wins, is a selected demonstration; preserve it separately and disclose outcome-driven selection. Proposed explanations and revised questions are new work, not established mechanisms. No uncertainty method or equivalence margin is supplied.

  Report an unmet practical target together with any positive observed magnitude rather than hiding either. The investigation is a fictional paper claim audit, not a real model trial.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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- Hypothesis expects B support strictly above A. Both pass 8/10 reserved cases. Show the intermediate steps and give the conclusion with its scope.

Main claim step	Complete result	Supported wording
A observed support passes	8/10 = 80%	Report supplied count and denominator
B observed support passes	8/10 = 80%	Same observed count
Observed difference	80% − 80% = 0 percentage points	Strict improvement not observed
Original expectation	B strictly above A	Expectation unmet on these cases
Population equality	Not tested or established	Do not say both methods are universally identical

Main claim step	Complete result	Supported wording
Next question	Can be proposed with a new design	Preserve original result and identify new exploratory work

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Negative results state that evidence did not meet a declared expectation, with the original method and limitations retained. Main prediction expects B support strictly above A; both pass 8/10 = 80%, so observed strict improvement is absent. This does not prove equal population performance or universal uselessness. Separate picture predicts higher B correctness but both score 6/8 = 75%.

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  Report an unmet practical target together with any positive observed magnitude rather than hiding either. The investigation is a fictional paper claim audit, not a real model trial.

- A report omits this qualification: Failure to meet a criterion or reject a null is not proof of exact **equivalence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Negative results state that evidence did not meet a declared expectation, with the original method and limitations retained. Main prediction expects B support strictly above A; both pass 8/10 = 80%, so observed strict improvement is absent. This does not prove equal population performance or universal uselessness. Separate picture predicts higher B correctness but both score 6/8 = 75%.

  Zero observed count difference is not an equivalence test, and individual correct/error identities are unknown. A new task chosen because A fails, on which B wins, is a selected demonstration; preserve it separately and disclose outcome-driven selection. Proposed explanations and revised questions are new work, not established mechanisms. No uncertainty method or equivalence margin is supplied.

  Report an unmet practical target together with any positive observed magnitude rather than hiding either. The investigation is a fictional paper claim audit, not a real model trial.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Hypothesis predicts mean latency reduction at least 2 seconds. Observed paired mean reduction .8. Report.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **negative result**s and **claim revision evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **negative result**s and **claim revision**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Negative result**s and **claim revision**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Negative result**s and **claim revision**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Hypothesis predicts mean latency reduction at least 2 seconds. Observed paired mean reduction .8. Report.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

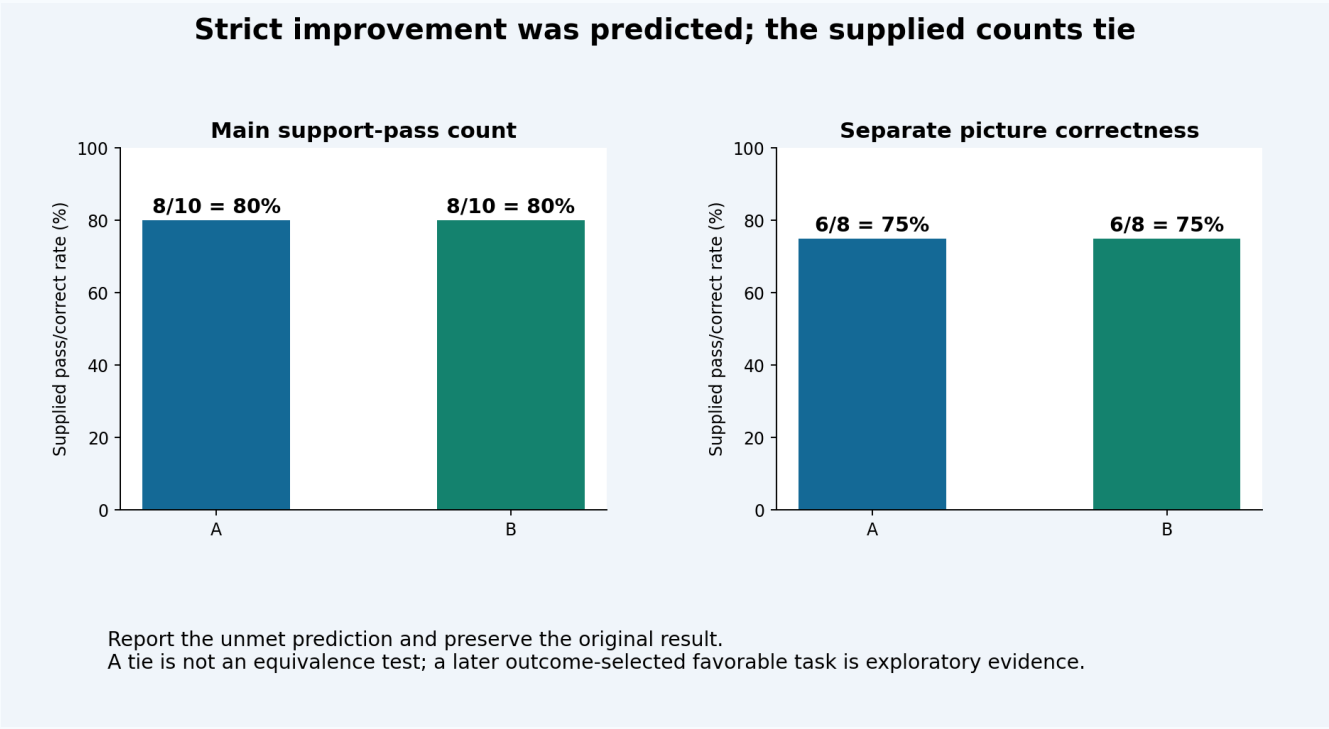
> Keep this task's inputs,rule and response together.

Complete task and evidence

- A **negative result** changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.
- A and B each score 6/8. Before running, the team predicted B would score higher.
- Does a 6/8 tie support the **prediction** that B scores higher? Explain why one newly selected favourable demonstration does not replace the original result or prove **equivalence**.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

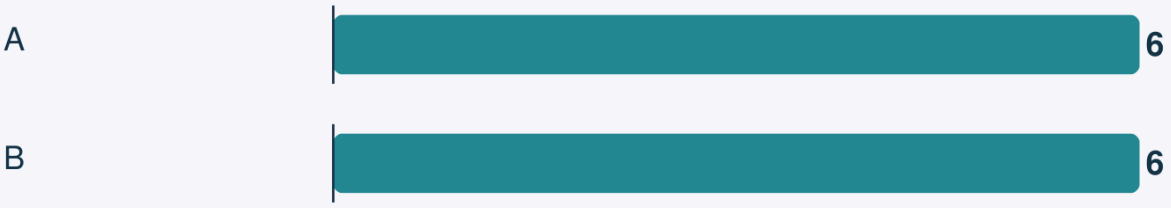
  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/28

No observed count advantage

correct cases out of 8 shared cases



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/28

Negative results and claim revision

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Recall the prediction

The claim was a higher correct count for B.



2. State the finding

Both versions score 75%; no observed count advantage appears here.



3. Revise cautiously

Report the unsupported prediction and a justified next test.

CHANGE ONE THING

B wins on one new task after the team chooses it because A fails.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A **negative result** changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.
- Which labelled step matches this detail? Report the unsupported **prediction** and a justified next test.
- 1. Revise cautiously
- 2. Recall the **prediction**
- 3. State the finding
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Revise cautiously Report the unsupported **prediction** and a justified next test. A **negative result** can be informative when methods and limits are clear.

A tie does not prove the methods are equivalent under all conditions or that the experiment was worthless.

Changed-case question and optional hint

- Changed-case practice: B wins on one new task after the team chooses it because A fails. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Treat that selected demonstration as exploratory, not a replacement for the original result.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Treat that selected demonstration as exploratory, not a replacement for the original result. Case selection now depends on outcomes.

Repair and retain the record

Review Robustness across conditions (ai-7/27). Goal: Explain the **mechanism** and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates.

Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Review Running the primary experiment (ai-7/25). Goal: Explain the **mechanism** and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Review Pilot experiments (ai-7/22). Goal: Explain the **mechanism** and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

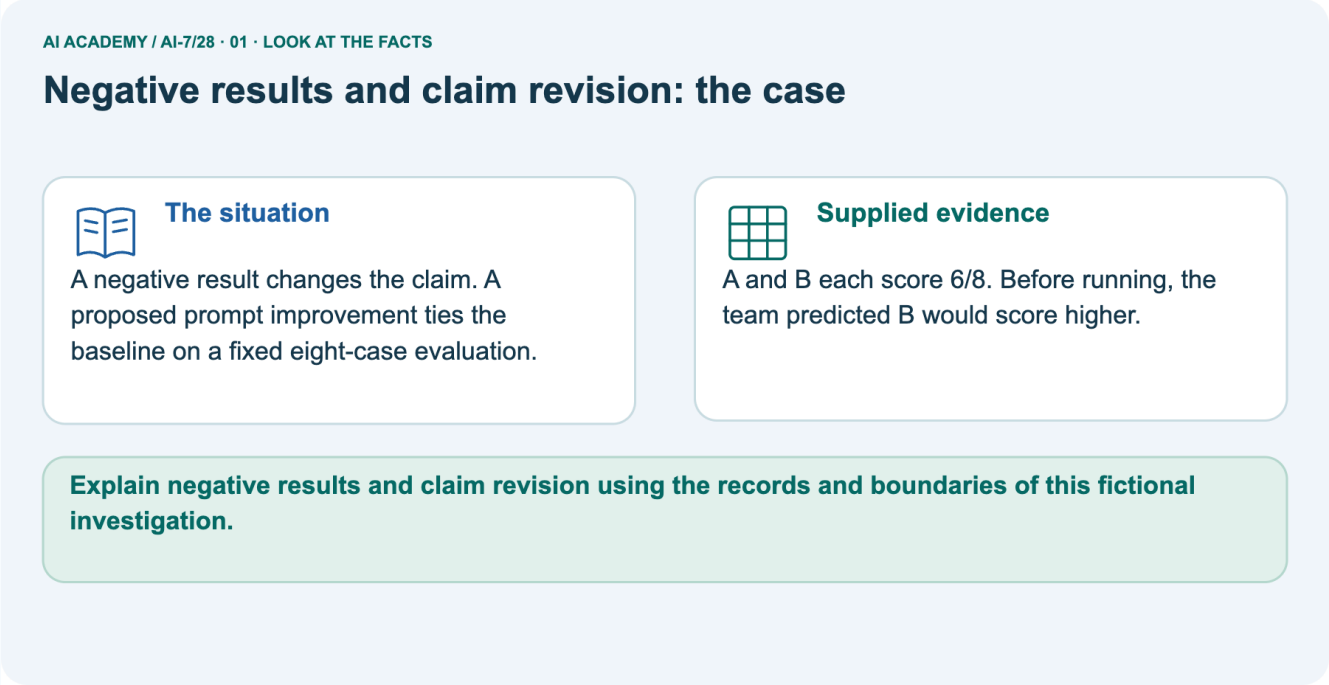
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Negative results and claim revision: the case



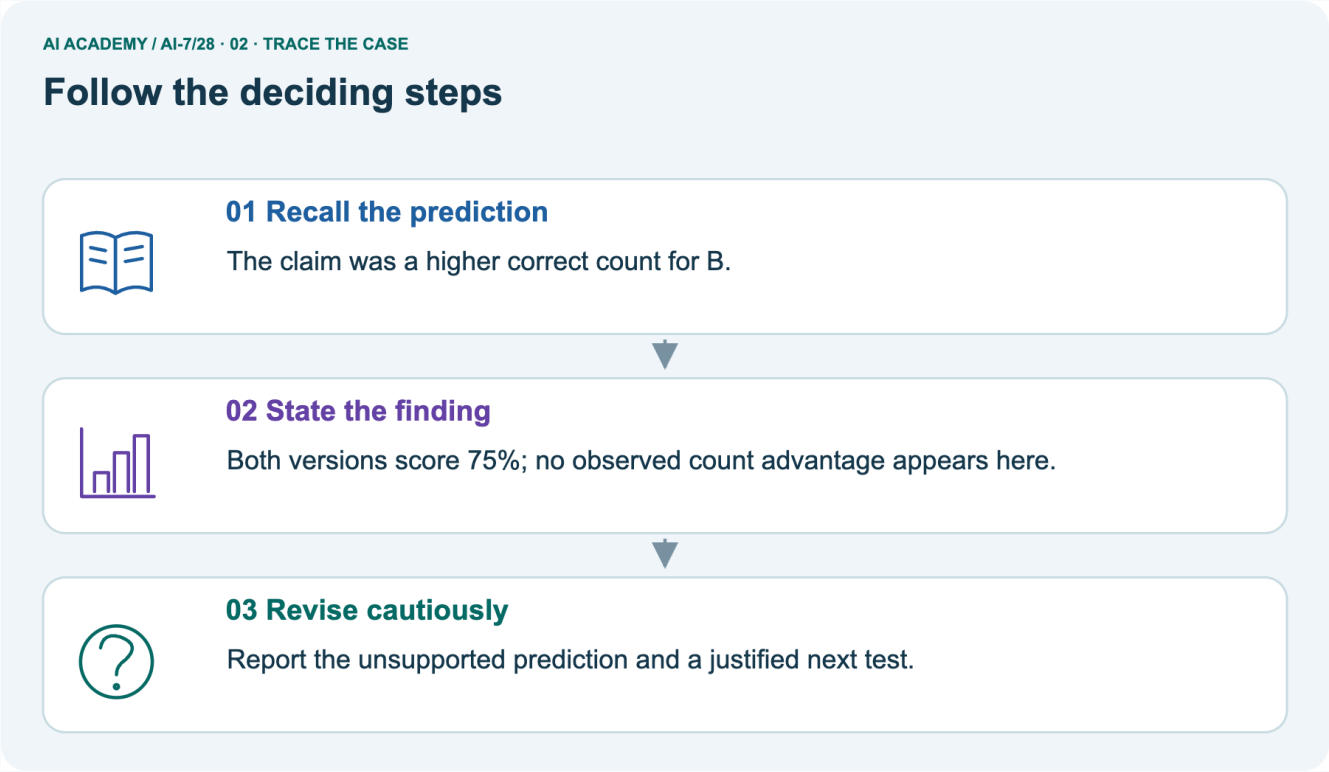
Follow the labelled picture with the matching explanation. Open the vector diagram

A **negative result** changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation. Supplied facts: A and B each score 6/8. Before running, the team predicted B would score higher.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



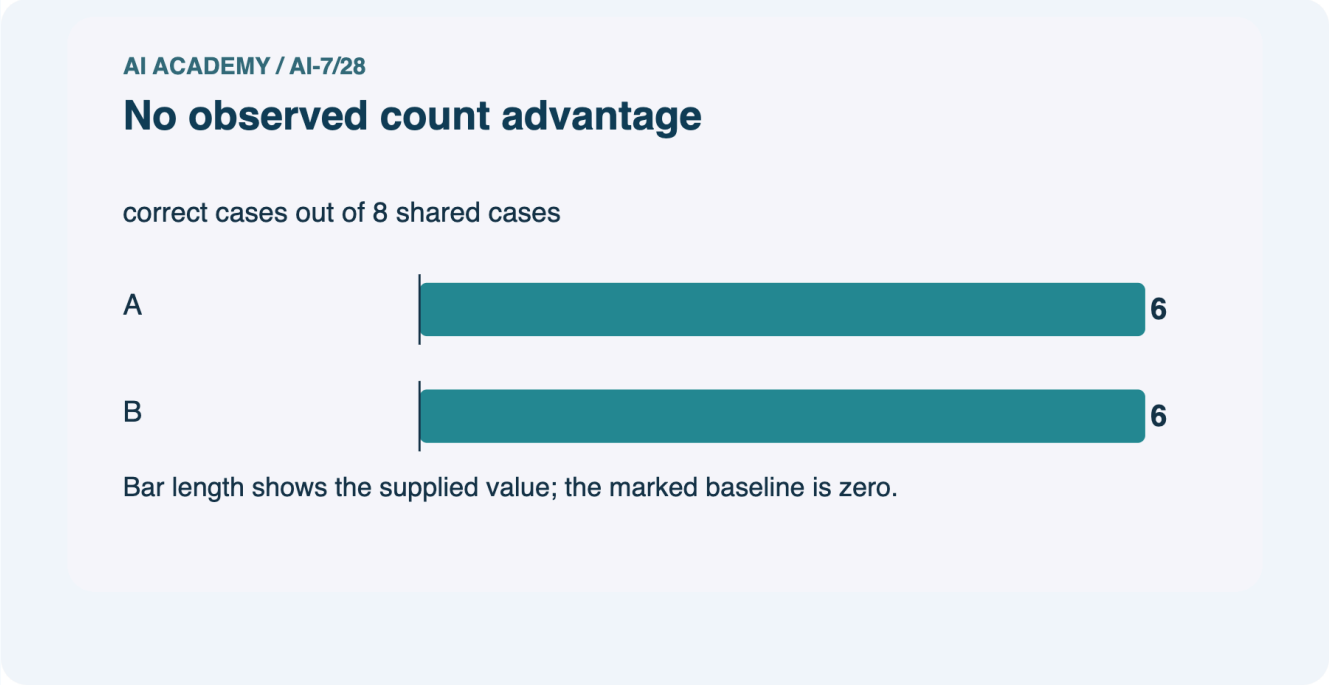
Follow the labelled picture with the matching explanation. Open the vector diagram

Recall the **prediction**: The claim was a higher correct count for B. → State the finding: Both versions score 75%; no observed count advantage appears here. → Revise cautiously: Report the unsupported **prediction** and a justified next test.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. No observed count advantage



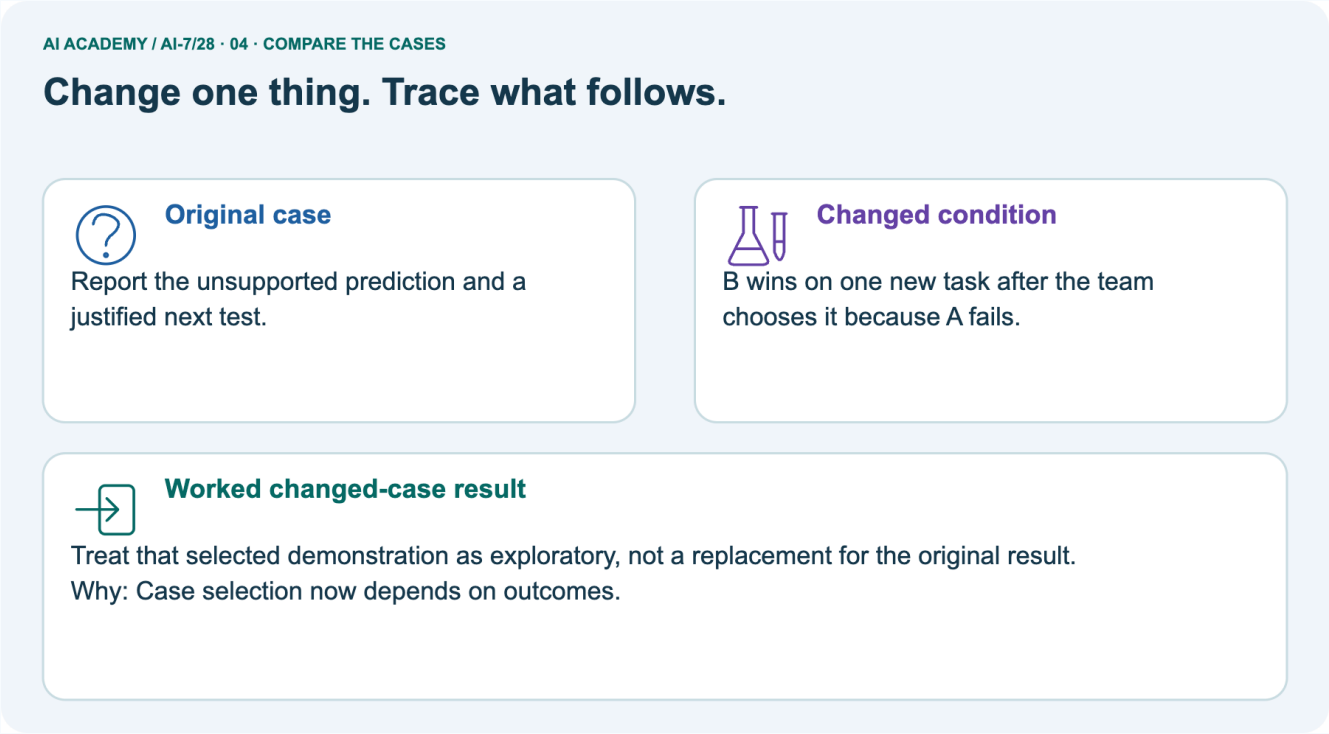
Follow the labelled picture with the matching explanation. Open the vector diagram

No observed count advantage. Units: correct cases out of 8 shared cases. A: 6. B: 6.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

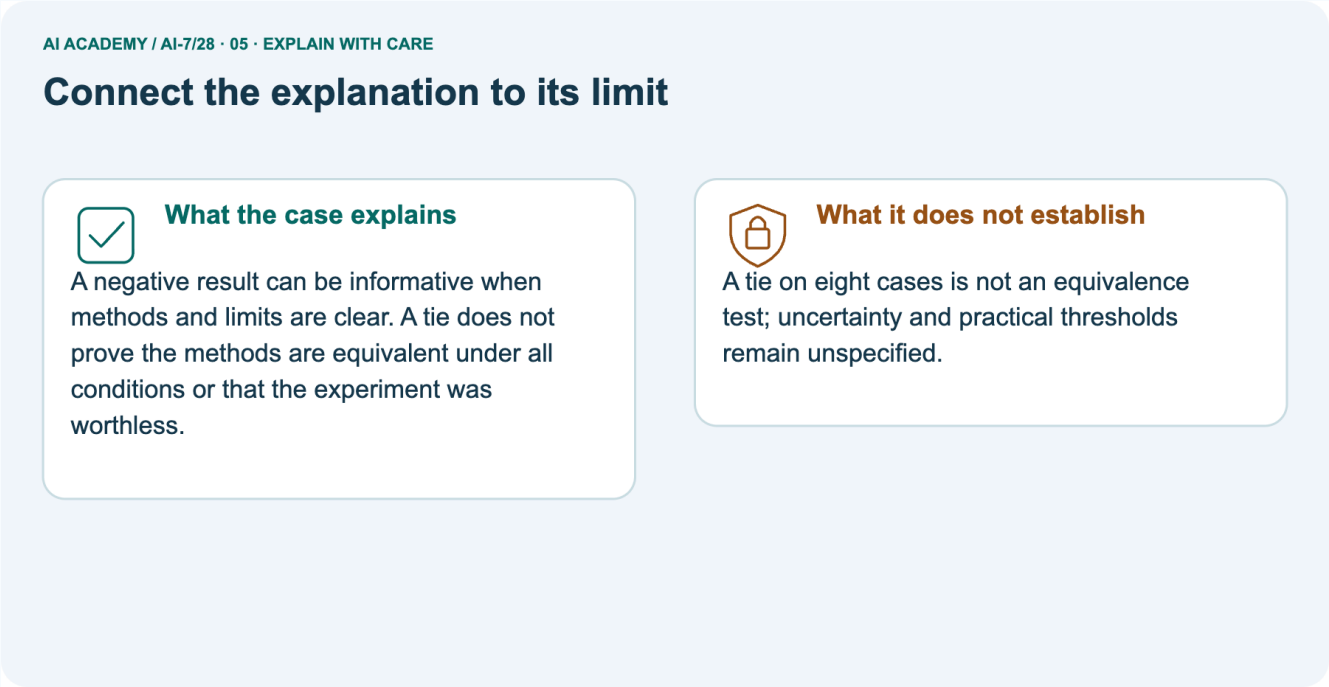
Original case: Report the unsupported **prediction** and a justified next test. Changed condition: B wins on one new task after the team chooses it because A fails.

Expected result: Treat that selected demonstration as exploratory, not a replacement for the original result. Reason: Case selection now depends on outcomes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

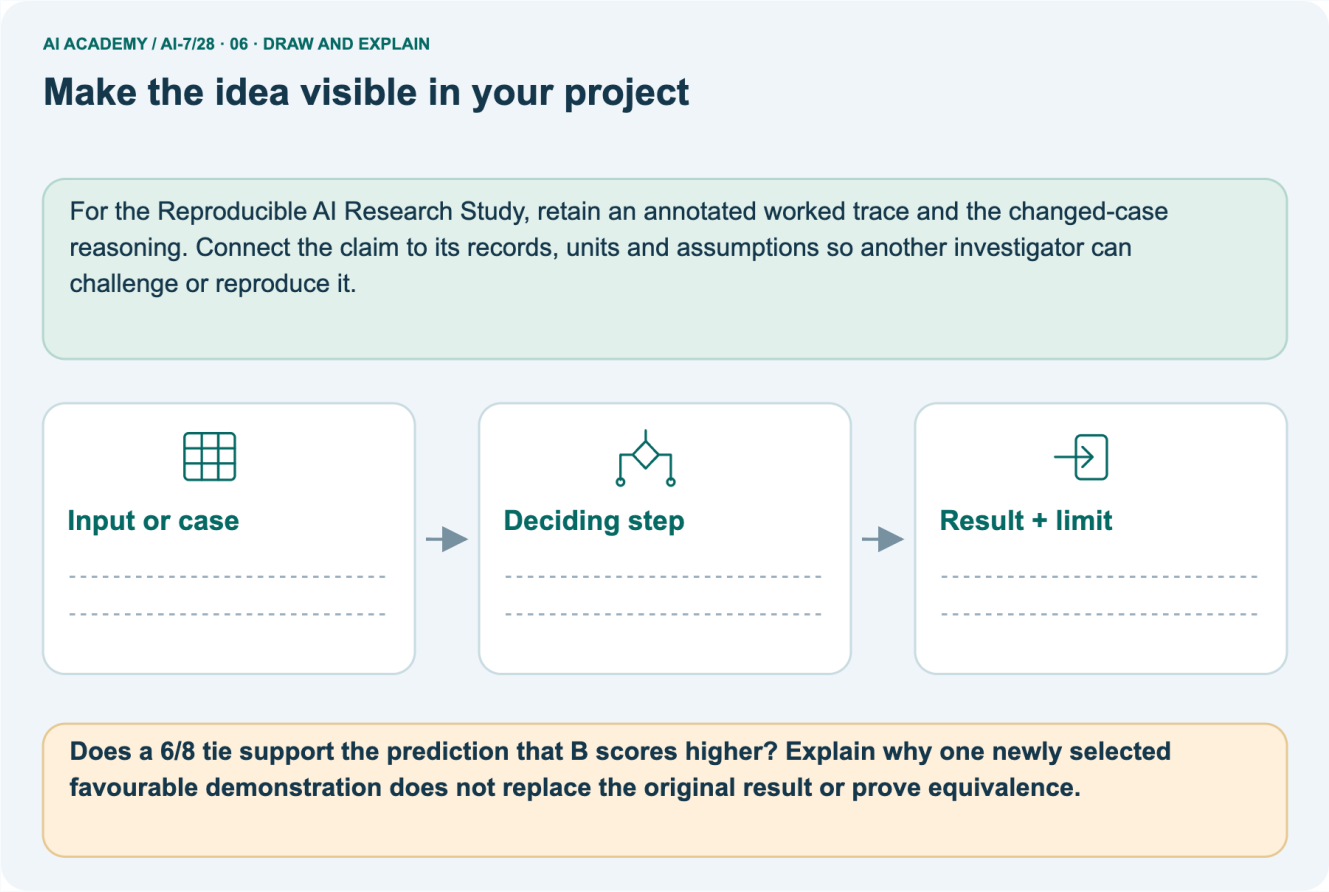
A **negative result** can be informative when methods and limits are clear. A tie does not prove the methods are equivalent under all conditions or that the experiment was worthless.

Limit: A tie on eight cases is not an **equivalence** test; uncertainty and practical thresholds remain unspecified.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Does a 6/8 tie support the **prediction** that B scores higher?

Explain why one newly selected favourable demonstration does not replace the original result or prove **equivalence**.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Negative results and claim revision: the case

02. Follow the deciding steps

03. No observed count advantage

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.